



(Don't Write anything in this Area)

- Q.7) (a) "Mendel's work laid the foundation of genetics, but human inheritance goes far beyond simple dominant and recessive traits." Discuss. (20 Marks)

Mendel law is based on the inheritance of a trait based on one gene, one character. He did this experiment using Monohybrid and Dihybrid cross over Pea plant.

He gave the four laws of inheritance

- (i) law of Unit factor : one gene = one trait
- (ii) law of dominance : dominant character will express in homozygous and heterozygous condition.
- (iii) law of segregation or purity of gametes.
- (iv) law of independent assortment
↳ each gamete will assort independently.

But the human inheritance goes beyond these two traits as:-

① Incomplete dominance :- In the various character, two alleles are not fully expressing itself

[Eg] A Red flower when mated with white the result is Pink (Mixture of both the flowers color.)

② Codominance : Both the allele express itself fully.

[Eg] In the ABO blood group, AB blood group is expressed by Codominance.

	I^A	I^O
I^B	$I^A I^B$	$I^B I^O$
I^B	$I^A I^B$	$I^B I^O$

③ Polygenic inheritance and multifactor inheritance $\therefore$ Various traits like height, weight colour are the result of various Alleles/genes contributing to a Common trait.

$\rightarrow$ In the population, there is a bell shaped curve

$\rightarrow$ It is based on the additive or cumulative affect of all genes.

$\rightarrow$ In population, medium/Median feature is selected more.

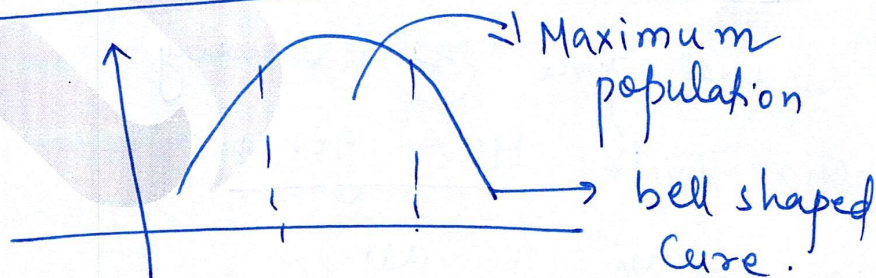
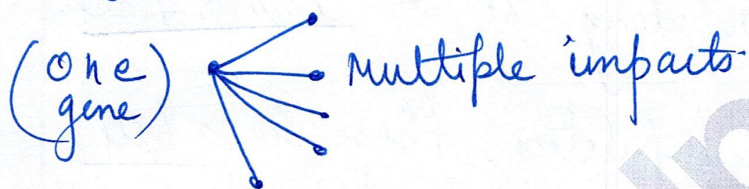


Fig: showing polygenic inheritance.

④ Pleiotrophy: Here the one gene is impacting several characters.

(eg) → Heart disease is caused by multiple genes and alleles.



⑤ Concept of genetic linkages is sometimes the genes are independently assorted and they create different kind of impact. So, here we can see that dominant and recessive trait have gone. (eg) usually happens with HCA genes.
6 genes with 110 alleles.

Thus, it is evident that traits in humans are shaped by multiple factors, not only dominant and recessive traits.